

SARAVANA KUMAR T

Data Analyst | ML Engineer | AI Developer

Chennai, India | +91-8940574870 | saravanakumar2751@gmail.com

PROFESSIONAL SUMMARY

Results-driven Data Analyst and ML Engineer with 4.5+ years of experience delivering data-driven solutions across fintech and development sectors. Proficient in Python, SQL, AWS, and Power BI. Proven track record in building machine learning models, automating workflows, architecting BI dashboards, and leading cross-functional teams.

TECHNICAL SKILLS

Languages & Querying: Python, MySQL, PostgreSQL

Cloud & Data Engineering: AWS (EC2, Redshift, Glue Jobs), DBeaver

Visualization & Reporting: Power BI, Streamlit, MS Excel, MS PowerPoint

Machine Learning & AI: EDA, Feature Engineering, Scikit-learn, Deep Learning, LangChain, CrewAI, Vertex AI

APIs & Integration: CIBIL API, CRIF API, Vertex AI API, Django, REST APIs, FAST APIs Bank Partner API Integration

PROFESSIONAL EXPERIENCE

Data Analyst | Muthoot Fincorp Ltd, Chennai

November 2024 – Present

DTCE – Digital Transformation & Customer Experience | Team Lead – UI Development

- Developed the Muthoot Fincorp Banking Assistant API utilizing Vertex AI (Gemini 2.5 Flash) and FastAPI to deploy a conversational chatbot that delivers highly concise, automated customer account summaries.
- Developed a Streamlit application (Pre-Login Check / BRE engine) leveraging CIBIL and CRIF APIs to automate credit eligibility checks, significantly reducing the TAT (Turnaround Time) from loan application login to sanction (Applied Patent in progress).
- Led the UI team within the DTCE unit, overseeing application development and data presentation standards across the department.
- Developed advanced chatbot Proof of Concepts (POCs) integrating LangChain with locally hosted LLMs (Ollama, Gemma); built custom Retrieval-Augmented Generation (RAG) pipelines to evaluate local inference limits, multi-agent logic, and prompt orchestration.
- Architected and built end-to-end dashboards for business stakeholders to monitor KPIs including credit performance, overdue rates, and sanction metrics.
- Conducted in-depth data analysis using Python (Jupyter Notebooks) for exploratory data analysis, feature engineering, and financial reporting.
- Managed multiple partner bank API integrations for automated data ingestion and processing.
- Engineered AWS pipelines using Redshift (stored procedures), EC2, and Glue Jobs for mail automation and job scheduling.
- Handled large-scale datasets (CIBIL tradelines) to generate financial insights that drive credit risk and business decisions.
- Built a machine learning model to predict customer loan-taking likelihood, enabling proactive customer targeting and improving conversion rates.
- Performed regular data quality monitoring and sanity checks to maintain data integrity across systems.

Key Achievements

- Won Project of the Year for Pre-Login Check Application – recognized for measurable impact on reducing loan processing TAT.
- Awarded Muthoot Blue Champion – internal recognition for outstanding contribution to digital transformation initiatives.

Project Executive | DHAN Foundation, Chennai

November 2020 – March 2024

Management Information System (MIS) | Managing 100+ Self Help Groups

- Monitored financial indicators of 100+ Self Help Groups using SQL queries, producing regular MIS reports.
- Generated analytical reports to assess performance indicators including overdue rates, NPA, credit linkages, and insurance metrics.
- Streamlined data reporting workflows reducing manual effort and improving accuracy of financial dashboards.

EDUCATION

B.Sc. Computer Science – First Class | Dr. Zakir Husain College

2015 – 2018

CERTIFICATIONS

- Foundation Level in Programming and Data Science – Indian Institute of Technology, Madras (IITM)
- Master Data Science Program – GUVI (IIT Madras)

KEY PROJECTS

Pre-Login Check Application (BRE Engine – Applied Patent)

Streamlit • CIBIL API • CRIF API • Python • AWS • Streamlit | Award: Project of the Year

- Automated credit eligibility validation at pre-login stage, drastically cutting loan sanction TAT and improving operational efficiency.
- Patent Number (**202641074634**) can be verified at - <https://iprsearch.ipindia.gov.in/PublicSearch/PublicationSearch/ApplicationStatus>

Customer Behaviour Prediction Model

Python • Scikit-learn • EDA • AWS EC2

- Built ML model predicting customer likelihood to take a loan using CIBIL tradeline data, enabling proactive sales targeting..